

Industrial Wrench Profile Optimization

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Problem Definition & Objectives

In this study, we design and optimize a thin plate with similar hexagon cutout that acts as a wrench. The part experiences a vertical concentrated load, providing a torque to tighten hardware located on the right hexagon cutout shown in Figure (1). The objectives of this design are to reduce the material as much as possible while maintaining a maximum effective under the yield stress of the material (163MPa), and below allowable overall deflection of 0.5mm. To better optimize the design of the wrench, another objective of this project is to better identify and reduce stress concentrations, as well as interpreting results based on element size and type to gain a more accurate understand of sufficient convergence and optimum method & sizing's. Evaluating results utilizing different meshing methods and element sizes gives a clearer depiction of the actual distribution of deflection and experienced stress, balancing allowable computational power and data accuracy. To fully optimize geometry given required loading, four improvements (including a base model) are designed to reach design specifications as previously discussed.

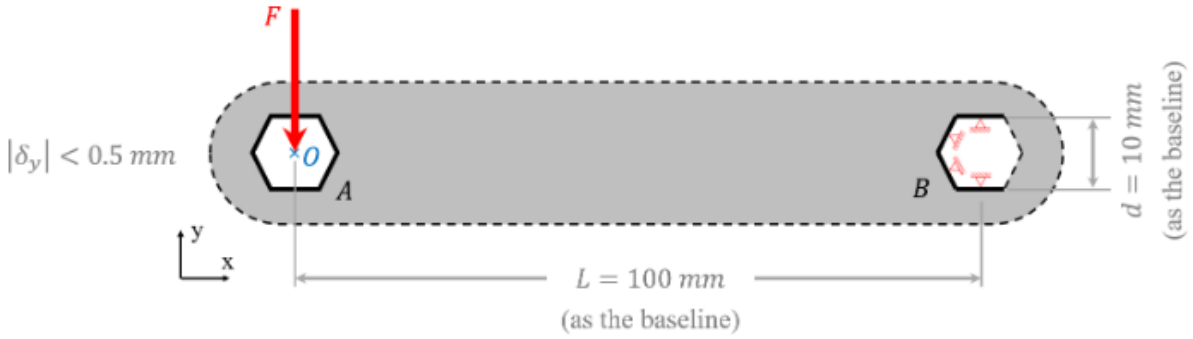


Figure 1: Part Schematic & Unmodified Dimensions

From the schematic provided above, and as explained previously, the thin plate is comprised of two similar hexagonal cutouts with a vertical concentrated load at the center of the left hexagon cutout. The right end, indicated by the fixed conditions, indicates the contact point of the wrench on the hardware component. The baseline hexagon spacing & overall hexagon height are listed in Figure (1), but is adjusted accordingly to the following parameters:

$$\underline{L = 92mm, d = 8.5mm}$$

To optimize the part geometry to minimize material while still being designed for allowable stresses, dashed borders can be modified through the design process. The black lines, however, as well as L and d, baseline dimensions provided above, must be maintained throughout geometry optimization. This is to ensure identical boundary conditions for experienced loading, as well as for the wrench itself to maintain correct function for it's specified use cases.

Modeling & Assumptions

As depicted in Figure (1), the wrench experiences displaced vertical loading at the center of the left hexagon cutout. This load produces torque that is applied to the jaw on the right end of the wrench, applying a counterclockwise moment, tightening hardware for respective hardware of specified sizing (8.5mm height). To analyse and correctly interpret results based on applied loading, various assumptions are made to simulate actual loading conditions.

Vertical loading is assumed to be applied perfectly linear on the wrench with a constant 150N of downward force. The force is not modified over time and does not act in the horizontal (x) axis or within the z-direction (in/out of the page). Additionally, the vertical load is assumed to be applied directly at the center of the hexagon cutout. In actual conditions, the actual loading may be applied at slightly inconsistent loading, or in different directions, causing loading that is not simulated during the experiment.

Another critical assumption when optimizing the overall shape of the wrench is the fixed boundary condition applied to the jaw surfaces in contact with hardware. Physically, this means that the jaw end of the wrench is assumed to be perfectly rigid and contain zero displacement at the end points. It is applied to simplify applied boundary conditions while effectively preventing rigid body motion of the component. In actual conditions, the jaw will open and rotate in small amounts to tighten hardware. By fixing the jaw end, worst-case shear and bending is applied to the jaw. In the context of analysis completed during this experiment, the wrench will be designed around worst-case loading and experienced stress at the jaw. To eliminate singularities that could be experienced through simulation results due to tight corners, specifically within the inner hexagon cutout edges, an inner edge is assumed to contain 1mm fillet radius. This also simulates actual conditions as wrench radii are generally given fillets on their edges to prevent fracture during machining processes and improved handling safety of hardware.

To model the wrench for efficient analysis, the part geometry was modeled using design modeler using a 2D sketch converted into a surface sketch with a prescribed thickness (3mm). In this context of analysis for this project, the thickness is assumed to be completely uniform and precisely 3mm on the entire body. This does not account for tolerances because of machining, creating small deviations in thickness, and resultingly, experienced stresses. In analysis, the part is assumed to be steel and contain the following material properties.

Youngs Modulus (GPa)	Poisson's Ration	Yield Stress (MPa)
210	0.3	163

The final assumption applied into analysis of the wrench is assumption of 2D plane stress within ANSYS. The software uses surface sketches from the part geometry with prescribed thickness to analyse 2D stresses within the model. Theoretically, this indicates that the state of stress contains one principal plane with zero experienced stress (Z).

Mesh Definition

To ensure accurate convergence and depiction of material optimization given required loading, the mesh has been defined to best balance computational efficiency and accuracy. For the base model and improvement cases, PLANE183 (Quadratic element order) has been applied. The results given using the linear element order (PLANE182) are provided for the fourth improvement case. PLANE183 ensures greater accuracy for improved design optimization for actual loading conditions. The quadrilateral dominant method is applied for analysis, ensuring high computational efficiency. To capture stress concentrations at edges and tight corners, refinement of 1 is applied on all applicable corners for each case. Finally, an element size of 0.5 is applied to the body, ensuring optimal computational efficiency, lowering convergence times. Images of applied methods, element sizes, and refinement settings, as well as a sample of meshing based on the final design are provided in Figures (2) – (6)

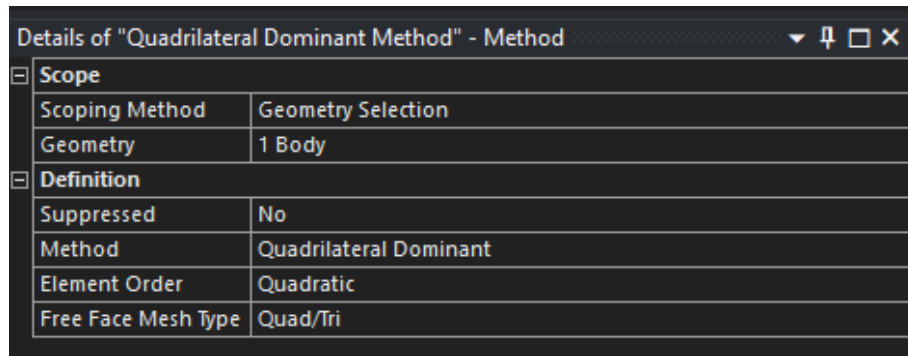


Figure 2: Method Details from Meshing

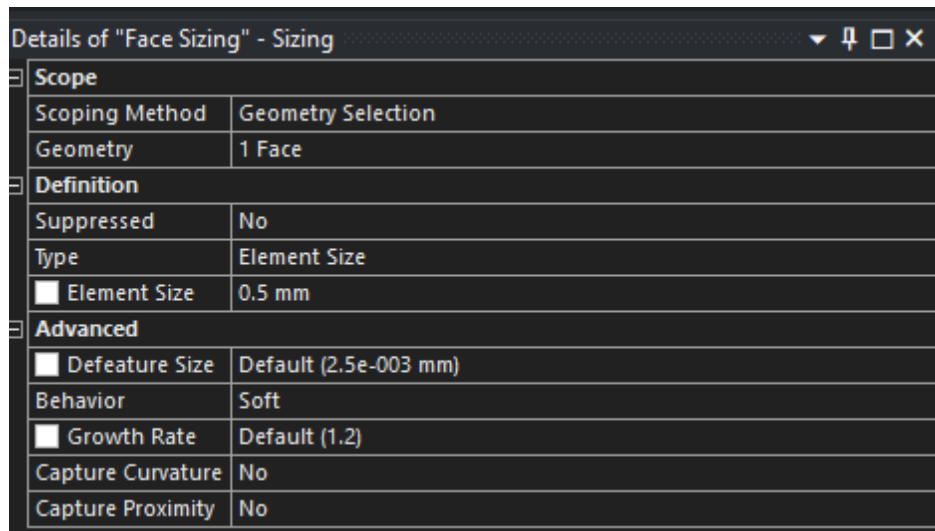


Figure 3: Face Sizing Details from Meshing

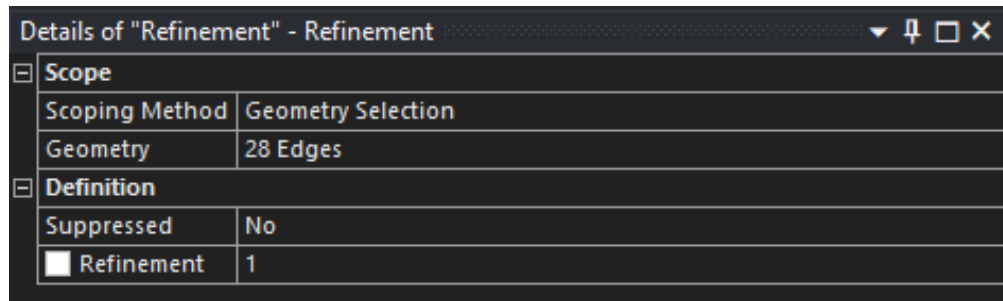


Figure 4: Refinement Details from Meshing

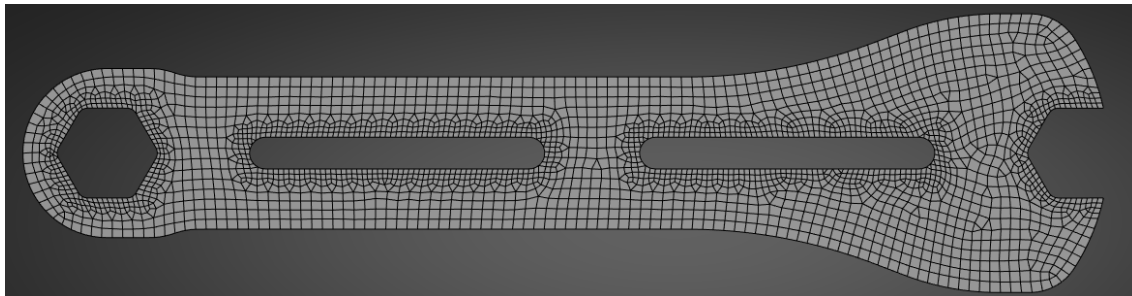


Figure 5: Sample Geometry Meshing Results with Refinement

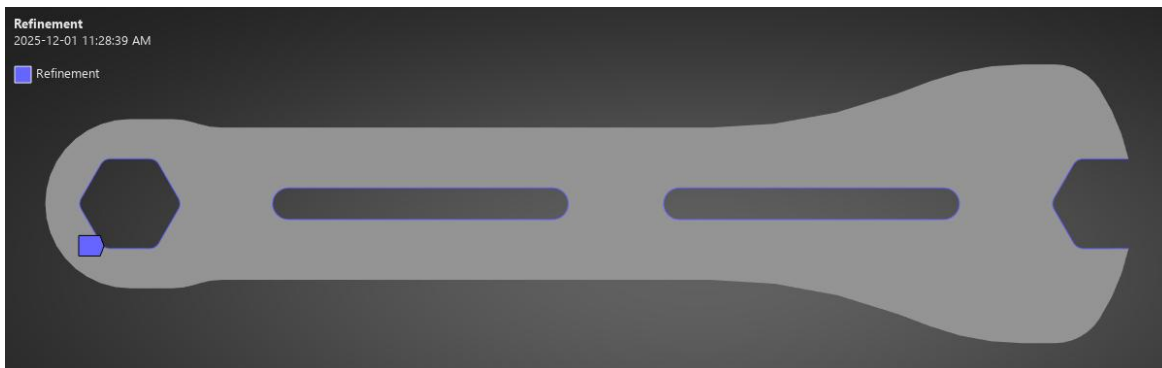


Figure 6: Sample Refinement Edge Selection

Boundary Conditions

As provided in previous sections, the wrench contains a fixed constraint located at the jaw on the surfaces shown in Figure (7). Additionally, the remote force (-150N) is applied on a remote position located at the center of the left hexagon cutout as shown in Figure (8).

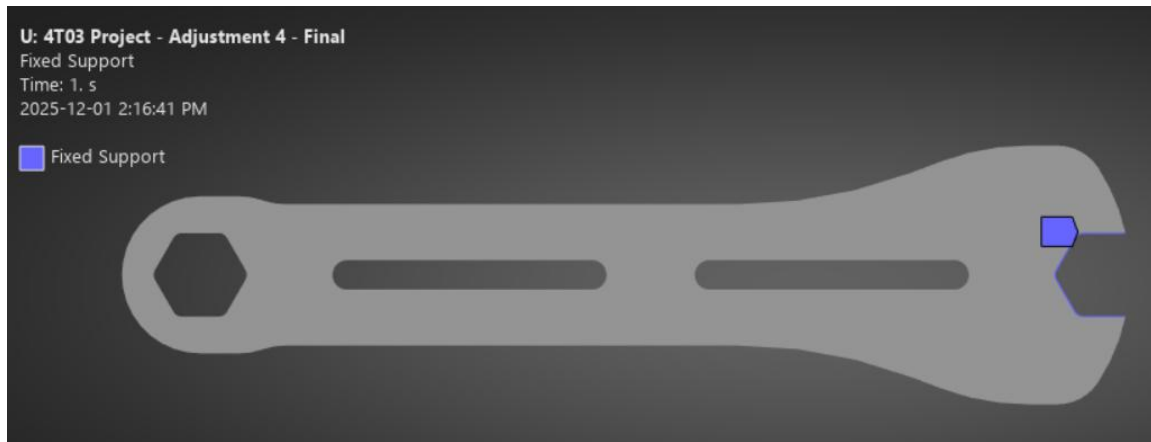


Figure 7: Fixed Boundary Conditions

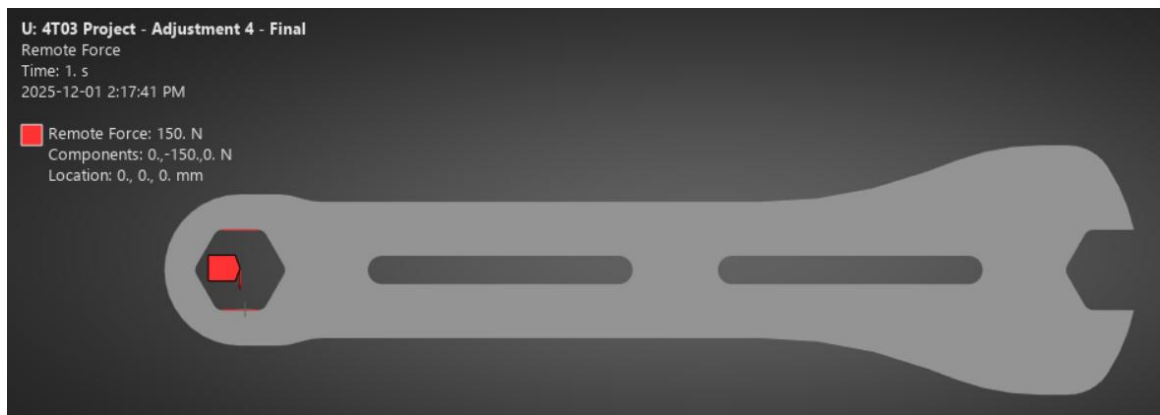


Figure 8: Applied Loading Conditions

Results & Discussion

Model 1: Non-Adjusted Design

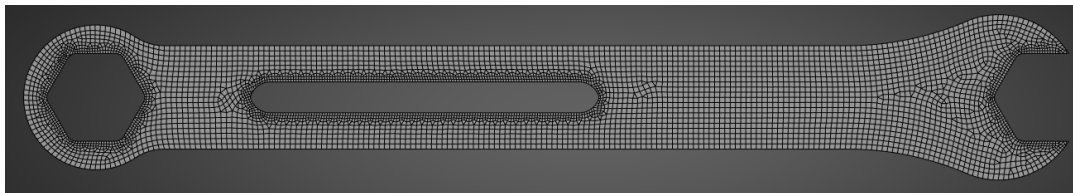


Figure 9: Non-Adjusted Design Geometry

The non-adjusted model contained the following results:

Volume (mm ³)	Number of Elements	Number of Nodes	Max Stress (MPa)	Max Deflection (mm)
2651.3	4967	15962	296.91	0.7565

As shown by resulting stress and deflection, both exceed allowable limits. The wrench fails in various locations, including both by stress concentrations at the inner cutout (Nodes 4774, 4540) and the

jaw (Nodes 14792, 15258), and by bending (Nodes 4469, 5227, 13569, 4705, 15652, 4944). Resultingly, this indicates that the part must be strengthened significantly at the long arm. Notably, more material can be removed at the center of the arm but must be removed carefully to not introduces new failure points by additional stress concentrations. This is removed in a future improvement. Additionally, the part must be strengthened at the jaw to reduce stress concentrations as well.

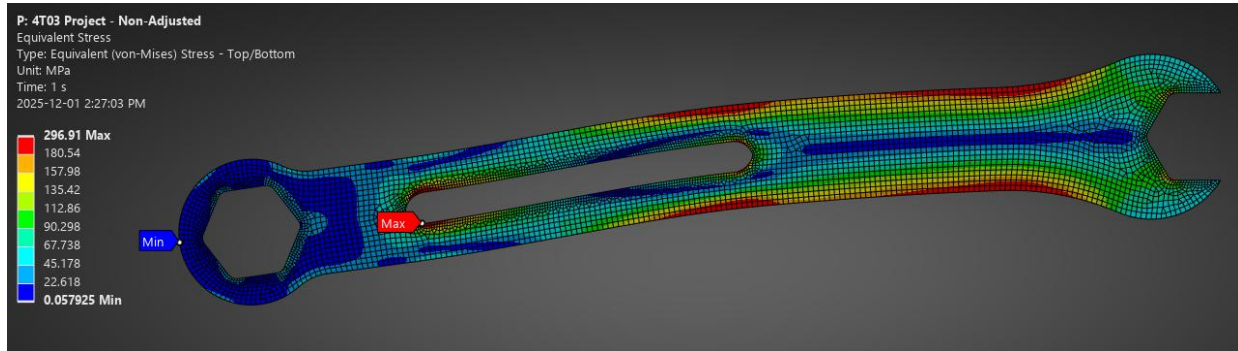


Figure 10: Study Results (Element Display) – Eq. Stress

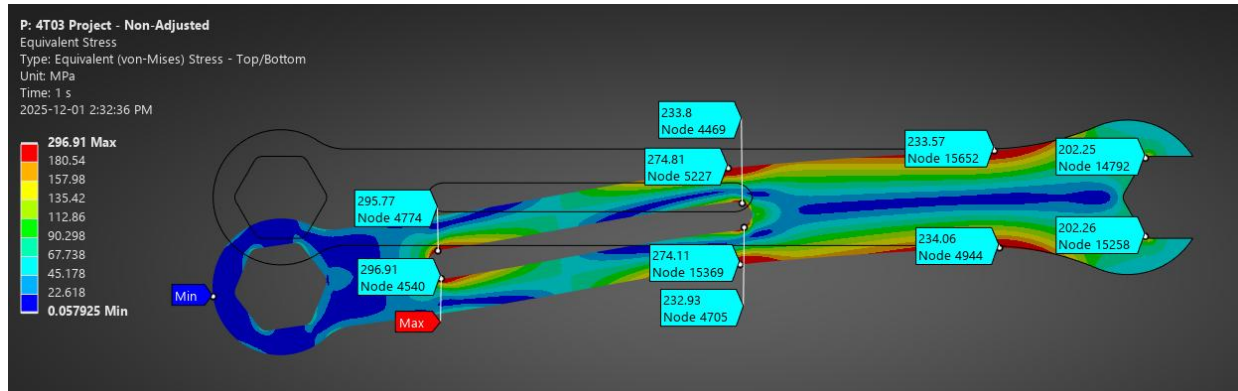


Figure 11: Study Results (Undeformed Wireframe) – Eq. Stress

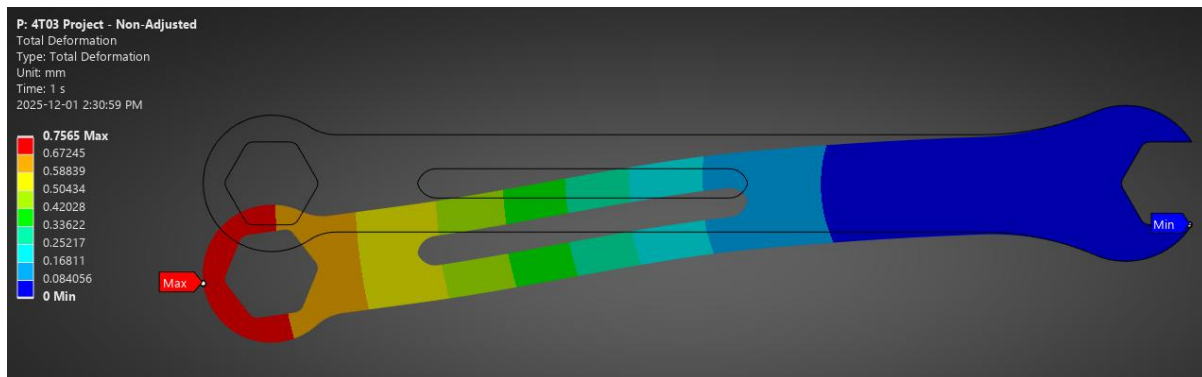


Figure 12: Study Results (Undeformed Wireframe) – Total Deformation

Model 2: Improvement 1 Design

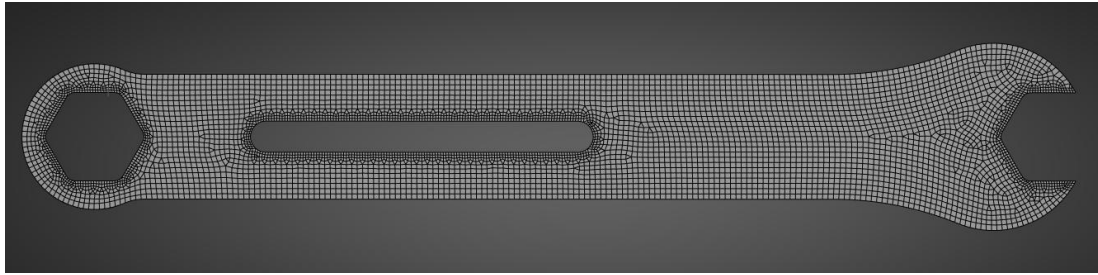


Figure 13: Improvement 1 Design Geometry

The first improvement design contained the following results:

Volume (mm ³)	Number of Elements	Number of Nodes	Max Stress (MPa)	Max Deflection (mm)
3202.2	5581	17524	206.18	0.43625

In the first improvement, the overall width of the wrench arm was increased, the slot slide was decreased, and the radius of the wrench at the jaw was slightly increased. Resultingly, the expected stress decreased by a large margin, where the max stress is experienced at the same location due to stress concentrations, but by approximately 100MPa less. Failure by bending is experienced at Node 5837 but avoids yielding at Node 17481. Stress concentration at the jaw produces stresses over yielding, as observed at Nodes 5320 and 5302. The overall deflection, however, is observed to be <0.5mm, which meets specifications for the design. Effort must be taken to strengthen the part around the jaw and to reduce stress concentrations at the slot, while effectively removing mass where possible.

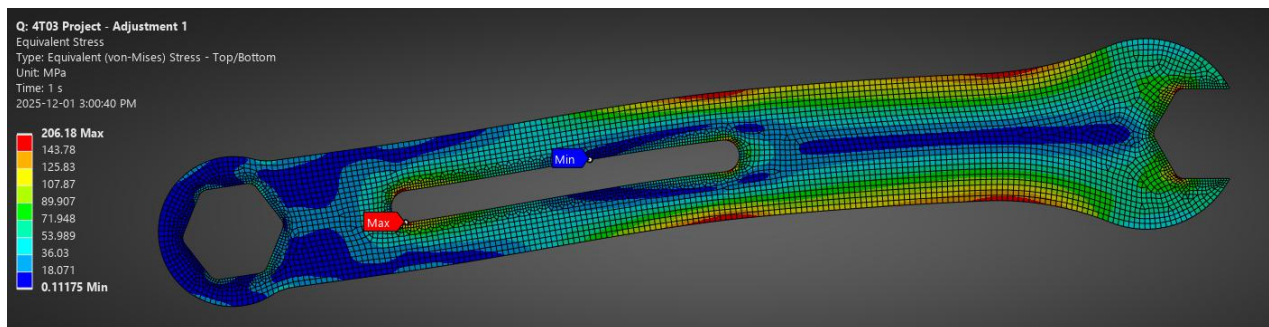


Figure 14: Study Results (Element Display) – Eq. Stress

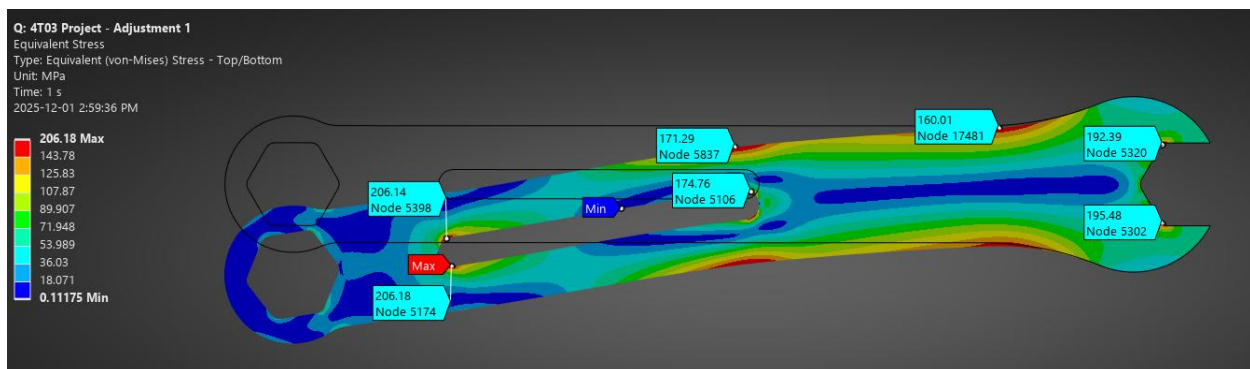


Figure 15: Study Results (Undeformed Wireframe) – Eq. Stress

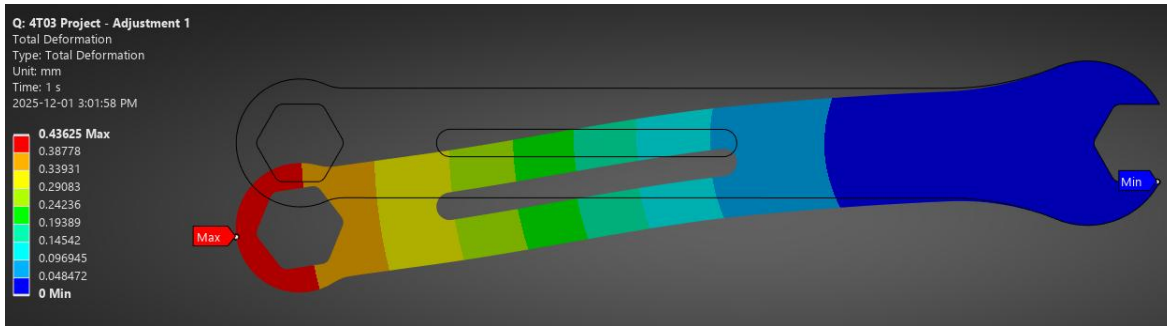


Figure 16: Study Results (Undeformed Wireframe) – Total Deformation

Model 3: Improvement 2 Design

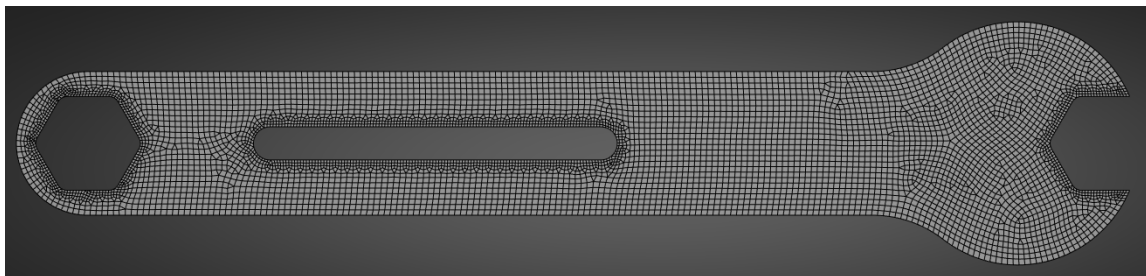


Figure 17: Improvement 2 Design Geometry

The second improvement design contained the following results:

Volume (mm ³)	Number of Elements	Number of Nodes	Max Stress (MPa)	Max Deflection (mm)
3588.1	6165	19198	178.55	0.32639

Geometry was modified from improvement increasing the width of the arm of the wrench and increasing the radius of the part around the jaw. The results from improvement 2 reduced overall stresses much less than the first improvement, where maximum stress within the part is now located at the upper and lower edges of the jaw (As seen in Nodes 6507 & 6489). The stresses at the slot also experience similar, yielding stresses (~168MPa). This indicates a focus on reducing effective stresses at the jaw. The stress concentration at the jaw is expected as the jaws are expected to be perfectly rid, creating high stress regions in its area as it tries to rotate because of the applied force.

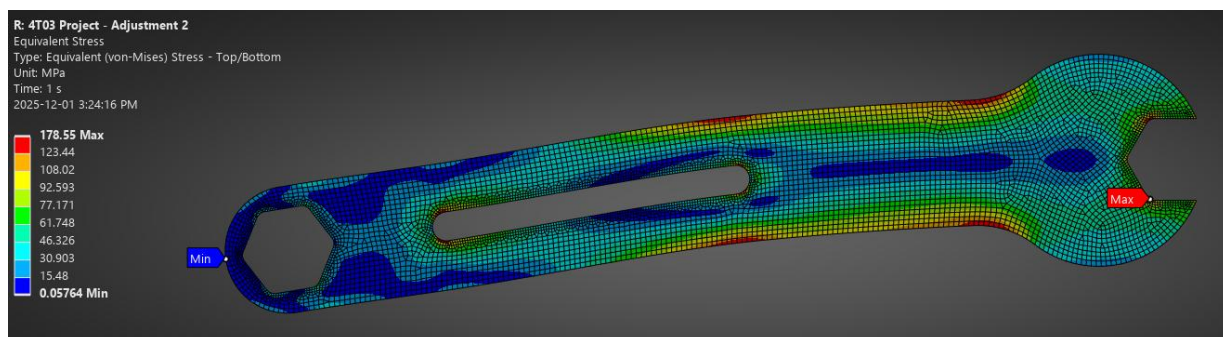


Figure 18: Study Results (Element Display) – Eq. Stress

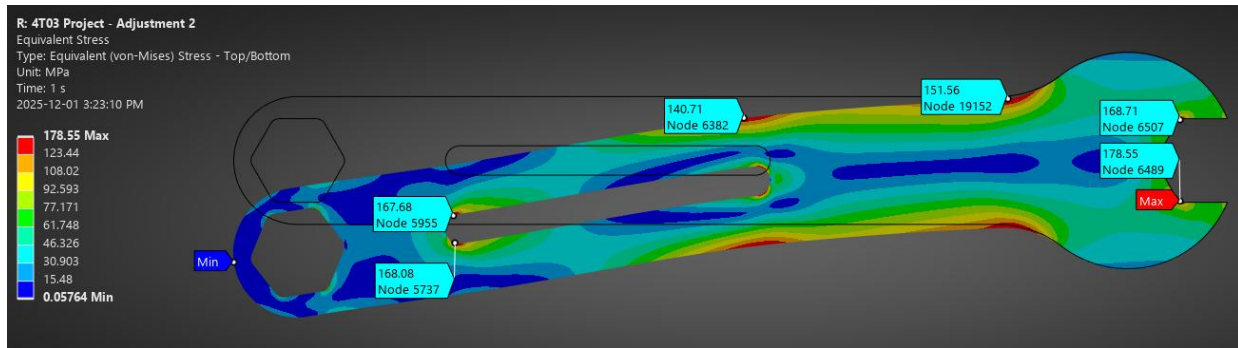


Figure 19: Study Results (Undeformed Wireframe) – Eq. Stress

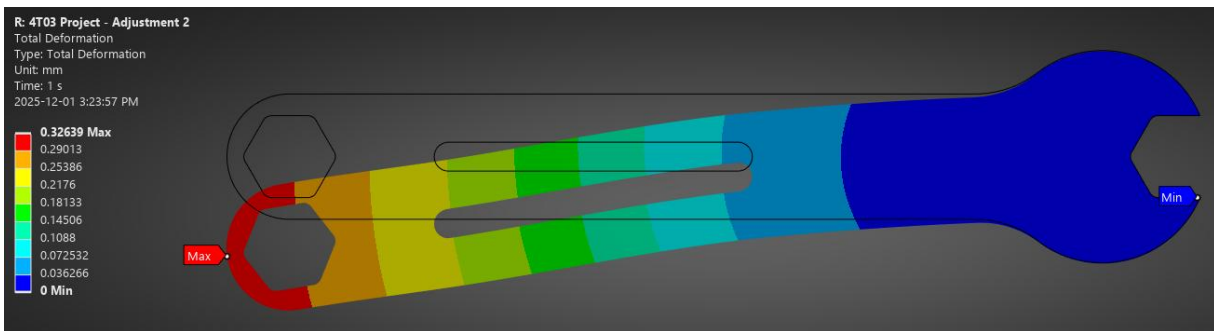


Figure 20: Study Results (Undeformed Wireframe) – Total Deformation

Model 4: Improvement 3 Design

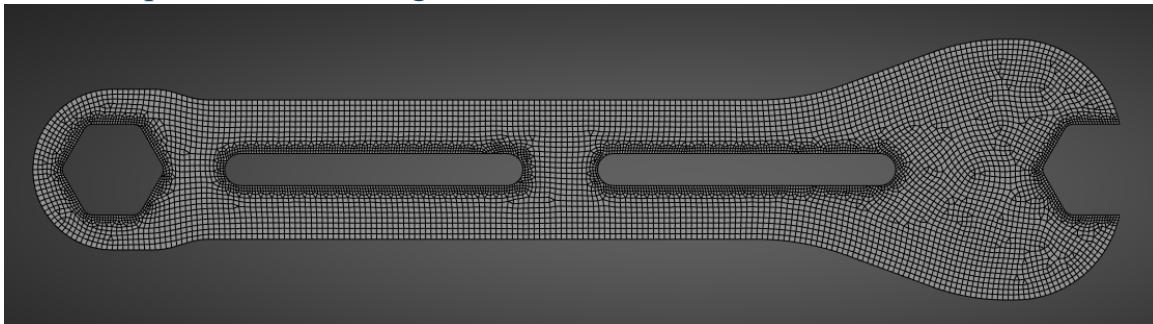


Figure 21: Improvement 3 Design Geometry

The third improvement design contained the following results:

Volume (mm ³)	Number of Elements	Number of Nodes	Max Stress (MPa)	Max Deflection (mm)
3786.6	7093	22215	169.94	0.26908

As observed from the table above and results below, the geometry was modified to include a second slot covering areas of low stress at the middle section of the arm of the wrench. This helped reduce the increase of volume because of the thickening of the overall wrench width at the arm and the jaw (primarily to reduce stresses at the jaw). The jaw was adjusted to have a smoother transition from the arm, reducing stress gradients at the transition. Additional, extra material was removed at the left side of the wrench, where low areas of effective stress are experienced. As seen below, all regions of high stress are well within allowable stress limits except at the jaw of the wrench. The stress experienced is very

close to the yield stress of the material, so smaller, optimized refinements to the material around the jaw must be made.

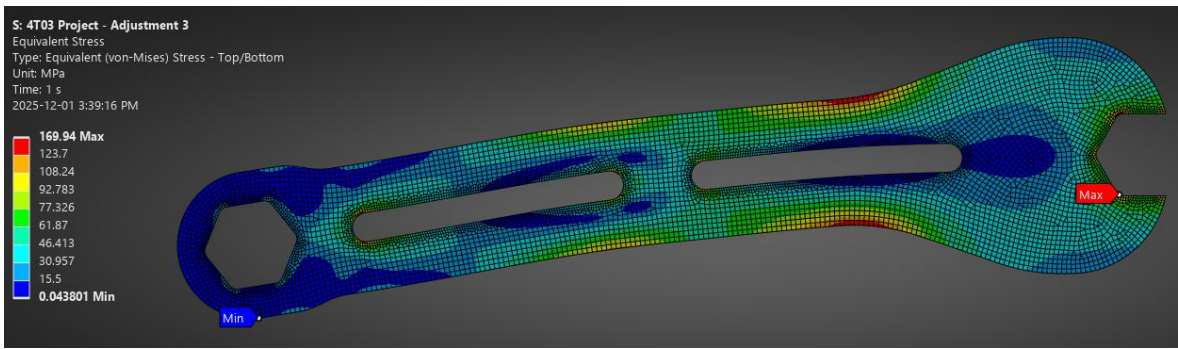


Figure 22: Study Results (Element Display) – Eq. Stress

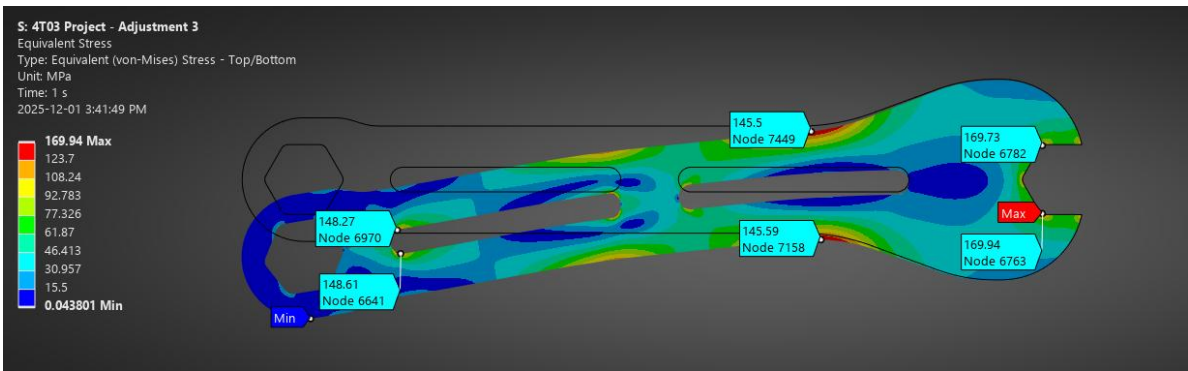


Figure 23: Study Results (Undeformed Wireframe) – Eq. Stress

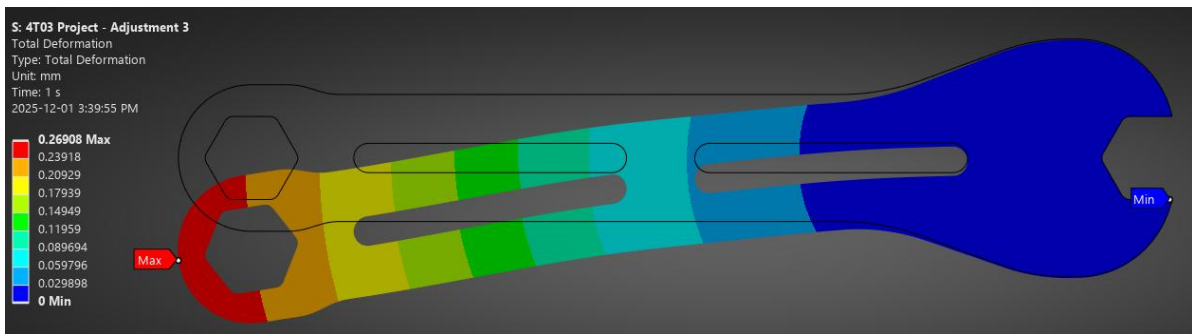


Figure 24: Study Results (Undeformed Wireframe) – Total Deformation

Model 5: Improvement 4 Design – PLANE 183

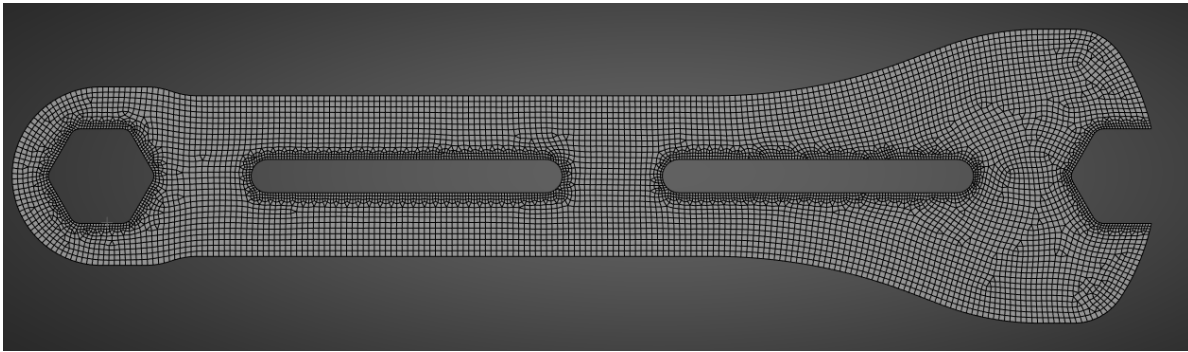


Figure 25: Improvement 4 Design Geometry

Specific Dimensions for the final improvement are provided below:

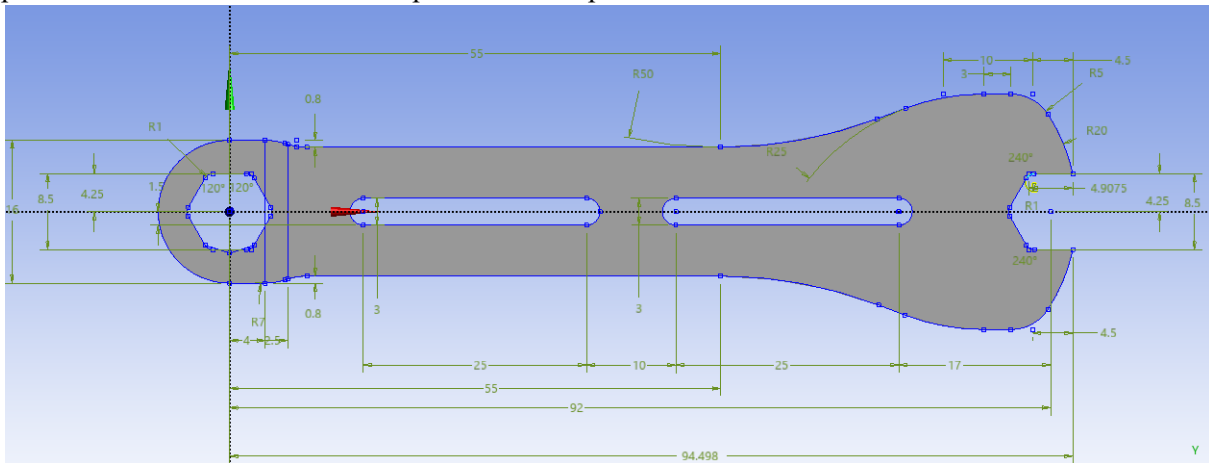


Figure 26: Improvement 4 Profile Geometry

The fourth and final improvement design contained the following results:

Volume (mm ³)	Number of Elements	Number of Nodes	Max Stress (MPa)	Max Deflection (mm)
4306.3	7939	24755	160.93	0.1983

The fourth iteration modified the third improvement by additional material at the jaw, and slightly increasing the width of the arm. The slots were also slightly moved towards the jaw, covering areas of low effective stress, and providing increased stress relief to the edges on the jaw where high stress concentrations are experienced. The resulting max stress at the jaw falls just below the yield stress of the material (Steel, 163MPa), indicating the design has been sufficiently designed. Note that the previous regions of high stress beside the inner jaw have been reduced significantly, as visible below. High stress concentrations at the inner jaw, based on the assumption of the fixed boundary condition, required a large deal of extra material to prevent failure. In actual conditions, where the jaw is not perfectly rigid, the design would be similar in size and geometry to Adjustment 2, where the slot would need to be modified slightly to ensure zero yielding. This design accounts for worst-case stresses at that location.

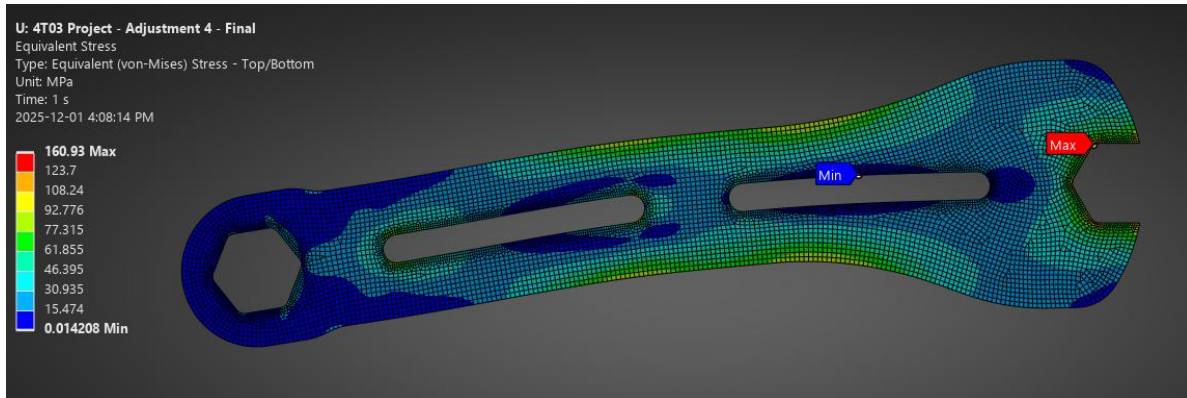


Figure 27: Study Results (Element Display) – Eq. Stress

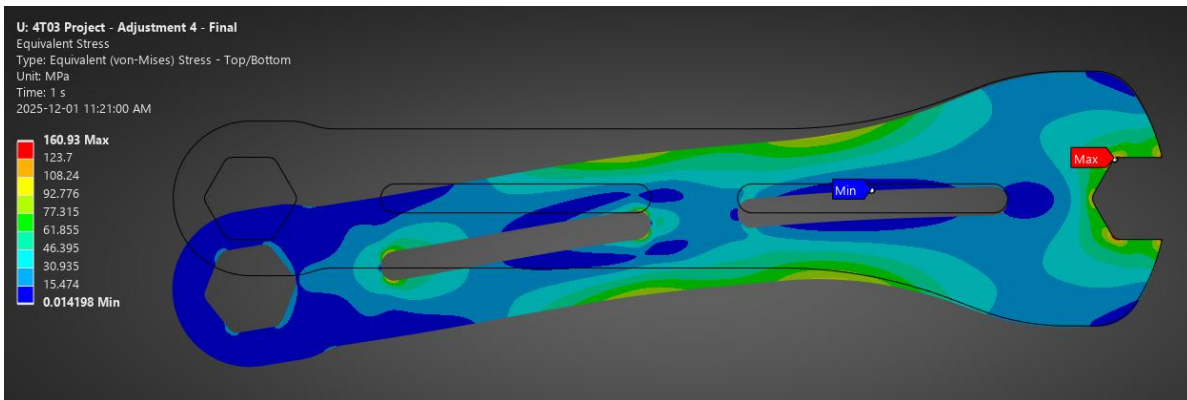


Figure 28: Study Results (Undeformed Wireframe) – Eq. Stress

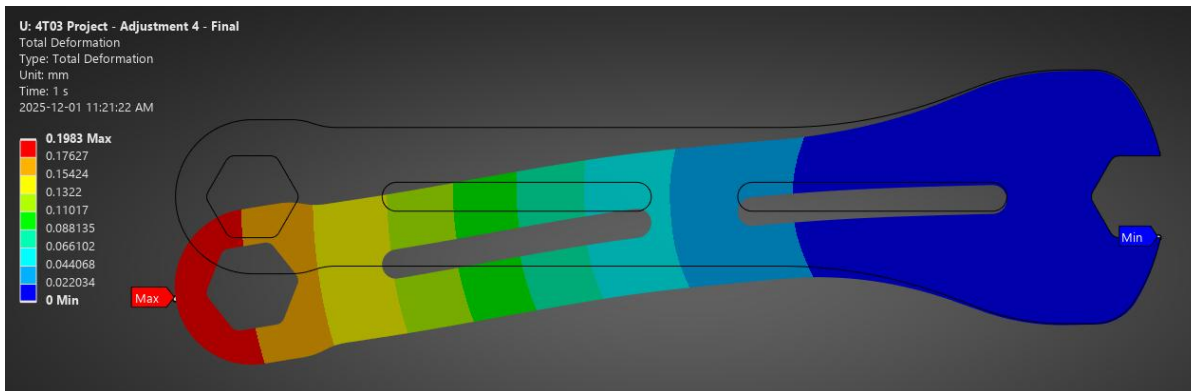


Figure 29: Study Results (Undeformed Wireframe) – Total Deformation

Model 5: Improvement 4 Design – PLANE 182

The fourth and final improvement design utilizing PLANE182 (linear element order) contained the following results:

	Number of Elements	Number of Nodes	Max Stress (MPa)	Max Deflection (mm)

PLANE182 Value	7692	8437	136.48	0.1986
% Change	3%	66%	15%	0.01%

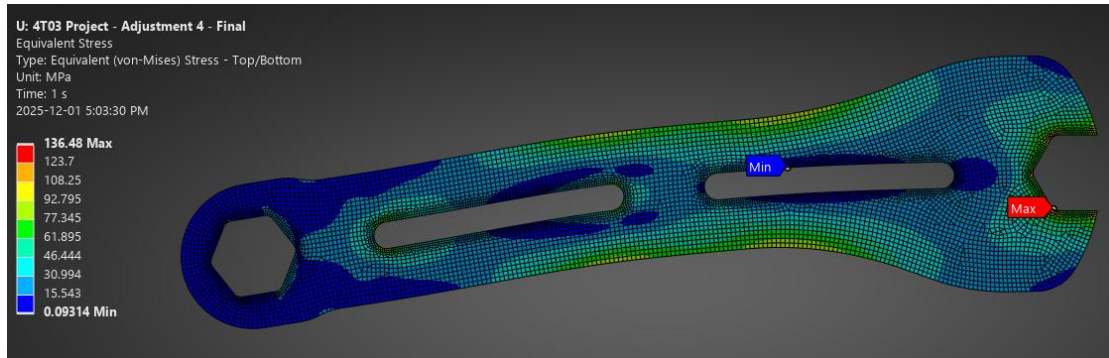


Figure 30: Study Results (Element Display) – Eq. Stress

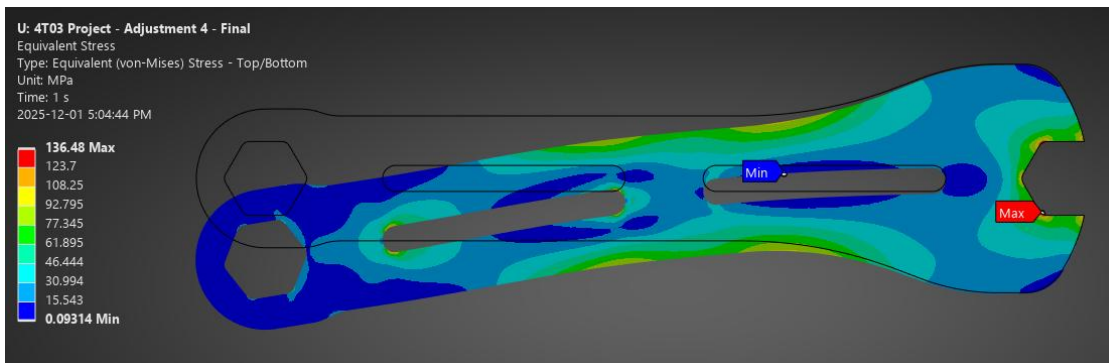


Figure 31: Study Results (Undeformed Wireframe) – Eq. Stress

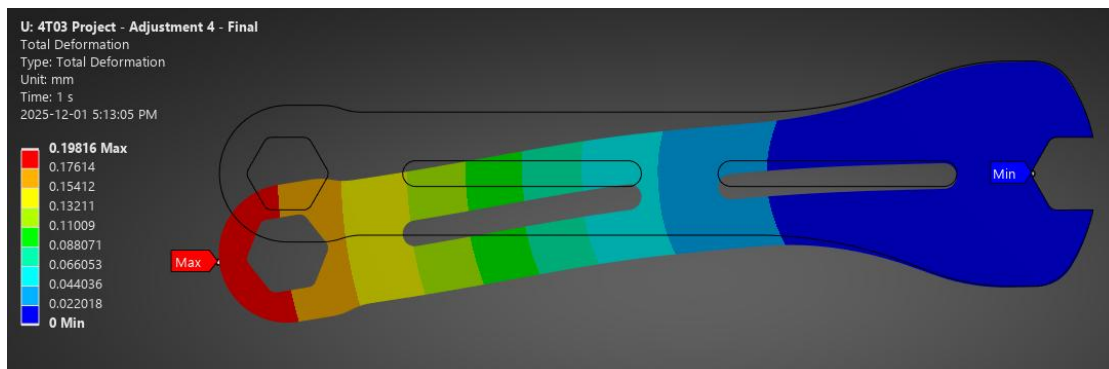


Figure 32: Study Results (Undeformed Wireframe) – Total Deformation

Elaborating on these results, it is observed that the number of nodes decreased significantly, differentiating 66% from the number of nodes found in the PLANE183 element order. Although the maximum deflection from the PLANE182 analysis differentiated very slightly from the total deflection from the PLANE183 analysis, it is found that the effective stress using the PLANE182 element order is approximately 33% less than in the PLANE133 case. This is because linear elements produce less accurate depictions of stress than the quadratic element order. In linear elements, the stress and strain remain constant, over the element, where an average stress is taken between the high peak of the stress of the element with surrounding stress values of the element. This “average” reduces equivalent stresses, as

seen in the previous study, where the quadratic order more accurately catches these stresses since the strain differs across the element, catching peaks within the element. In general, to gain a more accurate depiction of the mechanical behavior of components in loading, the quadratic element order will produce the most accurate results. If speed is of concern, linear element orders converge faster but result in inaccuracies such as those seen from previous studies.

Summary of Results

A summary of the results of the base model and improvement models are provided below. Each improvement decreases both the overall max stress and total deflection of the model. The results represent material was removed initially and strengthened in high-stress areas on each improvement. A clear trend, however, is found where the more material added (described quantitatively in volume) decreases the % change of overall stress and deflection. The main reason this occurs is due to the immediate strengthening completed in the first improvement before making smaller optimizations to reduce stresses at different regions. The first improvement helps strengthen various areas including stress concentrations at the inner slot, bending stress produced by the transition of the arm to the jaw of the wrench, and stress concentrations at the jaw as well.

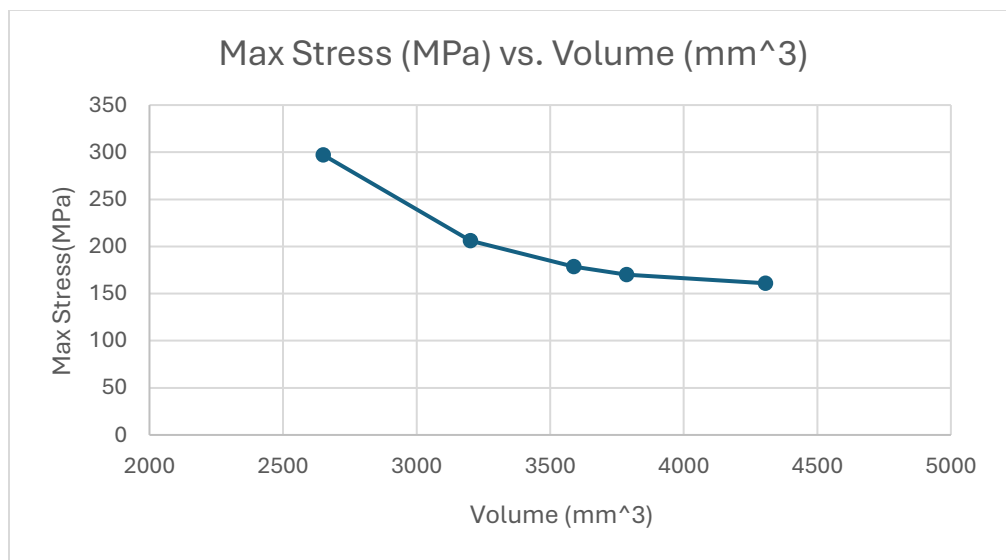


Figure 33: Max Stress vs. Volume (Overall Results)

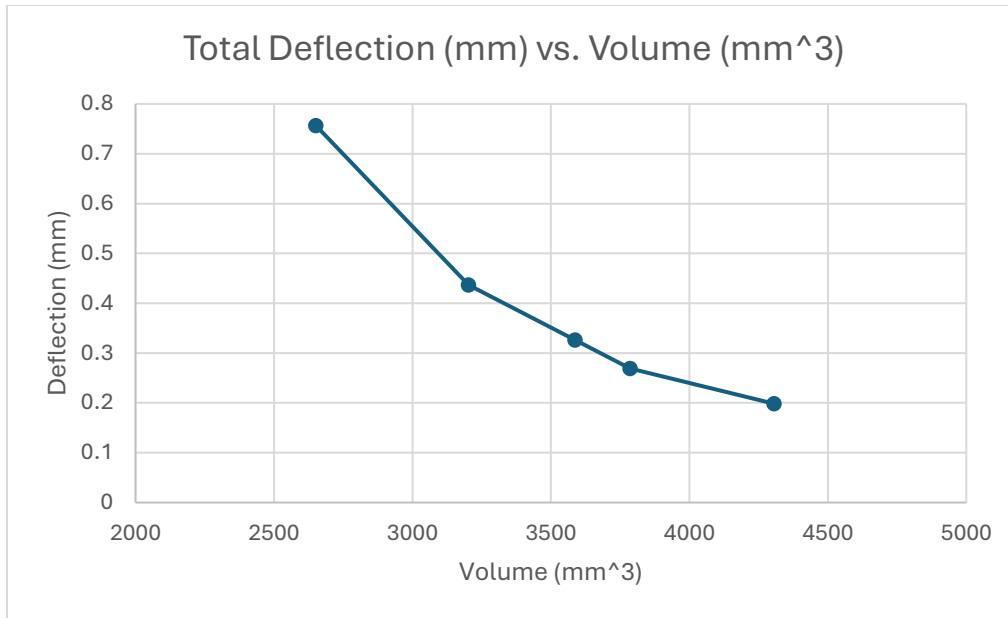


Figure 34: Total Deflection vs. Volume (Overall Results)

Mesh Sensitivity Analysis

A mesh sensitivity analysis was completed on the final improvement design using the PLANE183 element order (quadratic). When completing the analysis, refinements were disabled to gain a clearer depiction of stresses at smaller element sizes. When attempting to solve the simulation at element sizes lower than 0.4mm, convergence took long periods and was unable to converge utilizing refinements on meshing as previously completed for all improvements. To ensure consistency in data for all element sizes, and to gain a clearer picture of convergence, refinements were disabled, and a constant element size was applied to the face.

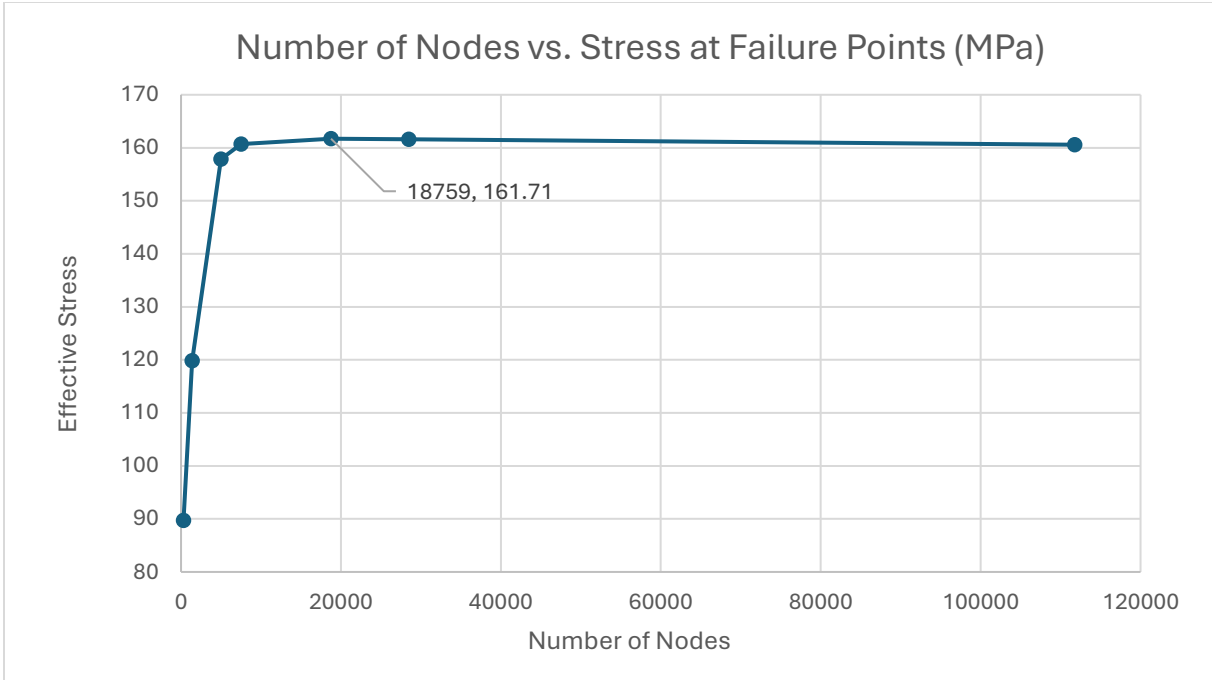


Figure 35: Mesh Sensitivity Analysis Results

The results of the mesh sensitivity analysis indicate that convergence occurs around 10000 nodes, (0.5-0.8mm), where the results begin to plateau at approximately 161MPa equivalent stress at the inner edges of the jaw. When selecting an appropriate element size for the application, one should ensure that the results have successfully converged, and can accurately depict regions of high stress, such as in high stress concentration areas with small radii. Thus, 0.5mm effectively balances lower computational power with accurate, converged results that correctly depict stresses in the region of failure within the study.

El. Size	# of Nodes	Stress @ Failure Points (MPa)
0.2mm	111787	160.58
0.4mm	28479	161.63
0.5mm	18759	161.71
0.8mm	7534	160.7
1mm	4992	157.87
2mm	1373	119.81
5mm	311	89.732

Conclusion

In this project, the design of a wrench was successfully optimized to reduce mass while maintaining the max experienced stress and deflection of the part under its respective material yield stress and allowable deflection. To optimize the design of the wrench, various improvements were taken to strengthen the part in high regions of experienced stress. More specifically, given the loading and geometry requirements of the wrench, effort was taken to reduce stresses primarily from stress concentrations and failure by bending. While the experience stress was lowered on each improvement, it also decreased total deflection, allowing both design parameters to be successfully improved.

Most significantly, the project explored the effects of assumptions and how setup parameters revolving around planar geometry, meshing, and the selection of boundary conditions contribute to results, and how those results should be interpreted with respect to loading in real-life conditions. For instance, as seen in later improvements, the selection of fixed boundary conditions at the inner jaw produced worst-case stresses, assuming rigidity that wouldn't be ideal in real-life situations. Although this may be beneficial to ensure sufficient design for any case, choosing less conservative boundary conditions (i.e. Allowing minimal rotation) would significantly reduce material usage, improving manufacturing costs while still being designed for actual behavior of the system. Similarly, to fully interpret the validity of these results, appropriate meshing must be applied to correctly visualize expected stresses, while maintaining sufficient computational performance to carry simulations accurately. Large meshing sizes struggle to visualize distributions on tight edges, leading to misinterpretations of results. This is evidently shown where the maximum stress experienced on larger element sizes were nearly twice as small as significant smaller element sizes when performing mesh sensitivity analysis. Element order also proved to produce significant deviations, as the linear element order study resulted in incorrectly lowering maximum experienced stresses. This could have led to insufficient strength within the design if the final design was found without using the quadratic element order. This project signified the importance of not just part geometric strengthening strategies but ensuring that the setup of your study matches actual expected behavior of the part in real life. Correct meshing and boundary conditions are critical to ensuring your part is sufficiently designed to avoid overdesign, contributing to unnecessary material costs, but meets design specifications to ensure specified performance and operational safety. As an engineer, it is your responsibility to make informed decisions on results from simulations that uphold to the standards of the engineering practice.